Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Cohen, Richard J. M.D., Ph.D.		
Project Title:	Effects of Simulated Microgravity on Cardiovascular Stability		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02142-1323	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	03/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	4	**No. of PhD Degrees:**	5
No. of PhD Candidates:	2	**No. of Master' Degrees:**	8
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Grenon, Marlene (Brigham and Women's Hospital)		
Grant/Contract No.:	NCC 9-58-CA00402		
Performance Goal No.:			
Performance Goal Text:			

The purpose of this investigation is to evaluate and test countermeasures to the development of microgravity induced orthostatic intolerance and susceptibility to ventricular dysrhythmias. We will apply the Cardiovascular System Identification (CSI) technique for assessing closed-loop cardiovascular regulation and Microvolt T-Wave Alternans (MTWA) technique for assessing susceptibility to ventricular dysrhythmias. The specific aims of the project are:
1. To test a panel of countermeasures to orthostatic intolerance in subjects intolerant to pre bed rest head-up tilt, in conjunction with the evaluation of CSI and other cardiovascular measures as a potential means of predicting orthostatic intolerance and countermeasure effectiveness.

2. To test in a 16 day head-down tilt bed rest study the effectiveness of the orthostatic intolerance countermeasure identified for each individual during pre bed rest screening, in conjunction with the evaluation of CSI and other

Task Description:	cardiovascular measures as a means of predicting orthostatic intolerance and countermeasure effectiveness. 3. To utilize MTWA analysis, in older men and post-menopausal women, to measure the effects of a panel of countermeasures for reducing cardiac electrical instability, and to study the effects of these countermeasures on baseline orthostatic tolerance and closed-loop cardiovascular regulation as measured by CSI and other cardiovascular measures. 4. To test utilizing MTWA analysis in a 16 day head-down tilt bed rest study the effectiveness of the cardiac electrical instability countermeasure identified for each individual during pre bed rest screening. 5. To further develop and enhance the methodology of CSI, MTWA and Plethysmography analyses.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has multiple earth based applications: Specifically, the research has direct application to the study of mechanisms of, and treatment for, orthostatic hypotension here on earth. Orthostatic hypotension is an an important and prevalent clinical problem which can lead to syncope and falls, which particularly in the elderly, may be life threatening. In addition, this study also focuses on mechanisms and treatment of life threatening ventricular dysrhythmias. Sudden cardiac death from ventricular dysrhythmias accounts for one-half of all cardiac deaths and one in seven of all deaths. The technologies developed in the context of this project all have direct application to earth based medicine. The microvolt T-wave alternans testing technique has already been successfully commercialized, cleared by the FDA, is reimbursed for by Medicare, and is in widespread clinical used to identify patients at high and low risk of sudden cardiac death sot that they may be appropriately treated. Just recently the Center for Medicare and Medicaid Services (CMS) announced an extremely broad proposed National Coverage Decision for microvolt T-wave alternans testing. National Coverage Decisions are limited to the 5-10% of covered procedures which CMS deems is of greatest importance to Medicare beneficiaries. This action by CMS demonstrates the broad national significance of this technology developed under NASA/NSBRI sponsorship and represents the contribution of space medicine to health care here on earth. Cardiovascular system identification is a technology that has potential widespread applications for a variety of disease states such as heart failure, hypertension, diabetes, etc which involve alterations in cardiovascular regulation. Continuous cardiac output monitoring is a technology that may have widespread clinical importance particularly in intensive care type situations. It may be particularly important for combat casualty care. Mathematical analysis of venous plethysmography data may play an important role in diagnosing mechanisms of orthostatic intolerance.
Task Progress:	• We have completed the initial studies in women and will finish the midrodine dose ranging study in them by March 25, 2006 (Specific Aim 2) We are currently enrolling other subjects into the study to complete Specific Aims 1 and 3. • We published five new manuscripts in peer reviewed journals and a sixth manuscript has been accepted for publication. • We have made presentations of our research at national and international meetings. • We have filed a second patent application on our continuous cardiac output monitoring technology. • The microvolt T-wave alternans (MTWA) technology developed under NASA/NSBRI support for the identification of patients at risk for sudden cardiac death is the subject of a very broad proposed National Coverage Decision by the U.S. Center for Medicare and Medicaid Services (CMS). MTWA is already covered by Medicare on a local carrier basis. National Coverage Decisions are reserved for only 5% - 10% of covered procedures which Medicare deems to be most critical for its beneficiaries.
Bibliography Type:	Description: (Last Updated: 01/18/2008)
Articles in Peer-reviewed Journals	Armoundas AA, Hohnloser SH, Ikeda T, Cohen RJ. "Can microvolt T-wave alternans testing reduce unnecessary defibrillator implantation?" Nat Clin Pract Cardiovasc Med. 2005 Oct;2(10):522-8. Review. PMID: 16186850 , Oct-2005
Articles in Peer-reviewed Journals	Grenon SM, Hurwitz S, Xiao X, Sheynberg N, Ramsdell CD, Kim C, Cohen RJ, Williams GH. "Readaptation from simulated microgravity as a stimulus for improved orthostatic tolerance: role of the renal, cardioendocrine, and cardiovascular systems." J Investig Med. 2005 Mar;53(2):82-91. PMID: 15810494 , Mar-2005
Articles in Peer-reviewed Journals	Grenon SM, Xiao X, Hurwitz S, Ramsdell CD, Sheynberg N, Kim C, Williams GH, Cohen RJ. "Simulated microgravity induces microvolt T wave alternans." Ann Noninvasive Electrocardiol. 2005 Jul;10(3):363-70. PMID: 16029389 , Jul-2005
Articles in Peer-reviewed Journals	Mukkamala R, Reisner AT, Hojman HM, Mark RG, Cohen RJ. "Continuous cardiac output monitoring by peripheral blood pressure waveform analysis." IEEE Trans Biomed Eng. 2006 Mar;53(3):459-67. PMID: 16532772 , Mar-2006
Articles in Peer-reviewed Journals	Grenon SM, Xiao X, Hurwitz S, Sheynberg N, Kim C, Seely EW, Cohen RJ, Williams GH. "Why is orthostatic tolerance lower in women than in men? Renal and cardiovascular responses to simulated microgravity and the role of midodrine." J Investig Med. 2006 May;54(4):180-90. PMID: 17152857 , May-2006
Articles in Peer-reviewed Journals	Xiao X, Grenon SM, Kim C, Sheynberg N, Hurwitz S, Williams GH, Cohen RJ. "Bed rest effects on human calf hemodynamics and orthostatic intolerance: a model-based analysis." Aviat Space Environ Med. 2005 Nov;76(11):1037-45. PMID: 16315396 , Nov-2005

Awards	"Aerospace Medicine Student and Resident Organization- Scientific Paper Award 2005." Jan-2006
Awards	"Association des Medicins de langue francaise du Canada- Recognition Prize 2005." Jan-2006
Patents	Mukkamala R, Sherman D, Cohen, RJ. "METHODS AND APPARATUS FOR DETERMINING CARDIAC OUTPUT." Jan-2006